Dolby MS12 V2 Release Note

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Revision History

Issue 0.6 (2023-06-21)

Update Dolby ASDK related.

Issue 0.5 (2022-10-25)

Update efuse burning command for new SoC

Issue 0.4 (2022-06-17)

Update some files location notes

Issue 0.3 (2021-06-02)

Modify the method of generating efuse pattern.

Issue 0.2 (2021-04-18)

Modify the method of ms12 decryption.

Issue 0.1 (2020-08-18)

This is the initial release.

Contents

1. Introduction

Amlogic Dolby MS12 IIDK implementation version is V2.4.1.

The newest Dolby MS12 SDK version for product certification is V2.6.x.

This document describes the release of the encrypted Dolby MS12 Library, including the Dolby efuse audio customer ID provisioning and verification after the efuse pattern burning is done.

2. Dolby Customer ID Provision

There is a unique Dolby customer ID needed to be written to efuse block.

There are two steps for efuse provision. The first step is to generate an efuse pattern before efuse is burned into the SoC. The second step is to burn the efuse pattern information to efuse in uboot.

2.1 Generating Audio ID efuse Pattern

If you want to generate audio ID efuse pattern, we provide an additional command to perform this. You can directly use the following command to generate the efuse pattern. Take the allocated audio id 0x13572468, T963/T963S (t5)series chip as an example.

```
$ cd bootloader/uboot-repo/

$ ./fip/audio_id_gen.sh --audio-id 0x13572468 --soc t5 -o audio_id.efuse
```

The audio ID entered here will be treated as a 32-bit number. If you want to input a hexadecimal number, you must add the prefix "0x", otherwise it will be treated as a decimal number.

Note

If you can not find this script or don't know the soc name of chip ,please ask for supporting from Amlogic.

2.2 Efuse Burning Provision in Uboot

It supports efuse burning provision in uboot, just use the corresponding command to burn the efuse pattern information to efuse. The efuse pattern file have two format for different soc, so need to check efuse pattern file first.

Chip supporting Plaintext format pattern：A5，S5，T5M，T3X

2.2.1 Ciphertext format pattern

If the efuse pattern file is a binary ciphertext format:

```
$ hexdump -C audio_id.efuse

00000000  2c bb 12 36 7e 5b 02 05  bf a6 3e 64 51 30 24 9b  |,..6~[....>dQ0$.|

00000010  6f af 2b 5d 9d db a1 d4  ff af af 33 fd 0d df 0a  |o.+].......3....|

00000020  59 00 58 2c ea 82 a7 33  a3 02 a7 9b 5e 4f f9 13  |Y.X,...3....^O..|

00000030  44 96 84 ce 9d 25 96 0b  f2 18 20 20 52 ce c2 8d  |D....%....  R...|

00000040  f6 98 48 d1 5f ec 73 cd  73 cd 9e 6c 21 6e f1 f9  |..H._.s.s..!n..|

00000050  df a7 fd a1 08 6d 40 cf  48 ab 13 b0 ec a7 af cf  |.....m@.H.......|

00000060  19 ff 22 5e ed 25 68 98  aa ee 3d 4e 51 08 7b 5d  |.."^.%h...=NQ.{]|

00000070  eb 3d 73 e5 97 24 2c d3  12 15 bf dc d4 b9 8b f9  |.=s..$,.........|
```

```
00000080   40 f1 70 e0 ae fc 20 07   07 1c ed 8b 1a d0 34 eb   |@.p... .......4.|
00000090   c1 3a ad 94 07 f2 22 c7   5f d8 1a 55 0f 2c f6 55   |.:....."._..U.,.U|
000000a0   91 5a 0f bf 02 52 8c ae   4c 66 d3 3b ba af 96 f0   |.Z...R..Lf.;....|
000000b0   37 3a 95 25 69 c1 88 ba   e2 49 83 e9 b3 12 25 0f   |7:.%i....I....%.|
000000c0   32 16 7a 02 39 90 95 94   47 6c f2 55 c0 3b a0 8e   |2.z.9...Gl.U.;..|
000000d0   68 d3 62 64 9e db eb 0a   db 2f 09 7f 1c be 72 6e   |h.bd...../....rn|
000000e0   78 58 12 cd da 33 a4 99   51 c0 25 60 0c 25 70 d3   |xX...3..Q.%`.%p.|
000000f0   c5 a9 e5 b4 db e5 8c e5   66 88 b6 30 b8 24 ac 99   |........f..0.$..|
......
```

You need to copy the efuse pattern file audio_id.efuse to the U disk and boot to the uboot console. Then you can use the following command for efuse burning.

```
#mw.b 12000000 0 400
#usb start
#fatload usb 0 12000000 audio_id.efuse
#md.b 12000000 400
#efuse amlogic_set 12000000
```

If you receive "aml log : Amlogic EFUSE pattern programming success!", it means EFUSE burning passed, otherwise it means it failed!

2.2.2 Plaintext format pattern

For some SoC, the efuse pattern file is the plaintext format, take the allocated audio id 0x13572468, A113X2 (a5)series chip as an example.

```
$ cd bootloader/uboot-repo/
$ ./fip/audio_id_gen.sh --audio-id 0x13572468 --soc a5 -o audio_id.efuse
$ hexdump -C audio_id.efuse
00000000   65 66 75 73 65 5f 6f 62   6a 20 73 65 74 20 41 55   |efuse_obj set AU|
00000010   44 49 4f 5f 56 45 4e 44   4f 52 5f 49 44 20 36 38   |DIO_VENDOR_ID 68|
00000020   32 34 35 37 31 33 0a 65   66 75 73 65 5f 6f 62 6a   |245713.efuse_obj|
00000030   20 6c 6f 63 6b 20 41 55   44 49 4f 5f 56 45 4e 44   | lock AUDIO_VEND|
00000040   4f 52 5f 49 44 0a                                   |OR_ID.|
00000046
$ cat audio_id.efuse
efuse_obj set AUDIO_VENDOR_ID 68245713
efuse_obj lock AUDIO_VENDOR_ID
```

You can see the audio_id.efuse is a plaintext file, the plaintext indicates some uboot commands, so you just copy the plaintext command to run in uboot console:

```
#efuse_obj set AUDIO_VENDOR_ID 68245713
```

```
#efuse_obj lock AUDIO_VENDOR_ID
```

Then if want to check the audio efuse burning status, just run the following command to check:

```
#efuse_obj get_lock AUDIO_VENDOR_ID
01
```

The command return “01”, it means efuse burning success!

Note

1, For plaintext in audio_id.efuse file. You MUST not change the plaintext in audio_id.efuse files by manually, and need to copy the plaintext command to run in uboot console directly.

2, Do not do the efuse burning provision with two different audio_id.efuse files which generated by two different audio id.It will cause the board can not be used anymore.

3. IIDK Related Library Reference

1. Released files lists

Efuse-audio-ID-of-customer.txt

libdolbyms12.so.customer

2. For Android project

Step 1 The encrypted Dolby MS12 library's name is libdolbyms12.so. please put the encrypted MS12 library at this folder: device/amlogic/common/dolby_ms12/install/encrypted_lib/libdolbyms12.so

Step 2 After the DUT is boot up, it will auto-decrypt the libdolbyms12.so(encrypted lib), and generate the decrypted file as /odm/lib/ms12/lidbolbyms12.so

Note

Here is the requirement of decryption:

- *The encrypted file is located in /odm/etc/ms12/libdolbyms12.so or /oem/libms12/libdolbyms12.so.*
- *The decryption tool is /vendor/bin/dolby_fw_dolbyms12*
- *The script file is "/vendor/etc/init/dolby_fw_dolbyms12.rc" or "/vendor/etc/init/dolby_fw_dolbyms12_oem.rc"*

Please confirm whether there are these files inside the DUT. If not, please check device/amlogic/common/Andorid.mk and add the platform name.

3. For Linux project

Step 1 The encrypted Dolby MS12 library's name is libdolbyms12.so. Please put the encrypted MS12 library at this folder:

multimedia/dolby_ms12_release/src/libdolbyms12.so

Step 2 After the DUT is boot up, it will auto-decrypt the libdolbyms12.so(encrypted lib),and generate the decrypted file as /vendor/lib/libdolbyms12.so(a soft-link of /tmp/ds/0x4d_0x5331_0x32.so)

Note

Here is the requirement of decryption:

- *The encrypted file is /usr/lib/libdolbyms12.so*
- *The decryption tool is /sbin/dolby_fw_dms12*
- *The script file is /etc/init.d/dms12.sh*

4. Confirm the Dolby MS12 library's version&git info(just use Android as an example):

Step 1 To play an DD+ file in APP(eg. Exoplayer).

Step 2 Capture the Log.

Note

- *12-17 00:03:36.310 16412 24099 I libdolbyms12: Dolby MS12 SIDK version:Version:2.4.0.b419a3a8*

- *12-17 00:03:36.310 16412 24099 I libdolbyms12: Dolby MS12 SIDK git version:b419a3a8af173c98317934c4cb6ec977796b300f*
- *12-17 00:03:36.310 16412 24099 I libdolbyms12: Dolby MS12 SIDK version serial:2040000b419a3a8*
- *12-17 00:03:36.310 16412 24099 I libdolbyms12: Dolby MS12 SIDK Last Changed:Date: Wed Nov 18 18:40:02 2020 +0800*
- *12-17 00:03:36.310 16412 24099 I libdolbyms12: Dolby MS12 SIDK Last Build: Wed Dec 16 11:08:57 CST 2020*
- *12-17 00:03:36.310 16412 24099 I libdolbyms12: Dolby MS12 SIDK Builer Name:Amlogic SH multi-media team*
- *12-17 00:03:36.310 16412 24099 I libdolbyms12: creating Dolby security session...*
- *12-17 00:03:36.320 16412 24099 I libdolbyms12: Amlogic - Dolby security recognize success!*
- *12-17 00:03:36.366 16412 24099 I libdolbyms12: Dolby MS12 enable,customer info AML-A-INFO:xxxxx*

If you have any question, please ask for supporting from Amlogic.

4. Android AIDK Support

Note: For android AOSP/Linux product, please ignore this chapter.

It is only available for Android 12 ATV project; others, please ignore this chapter.

Dolby introduced Dolby MS12 v2 Android Implementation Development Kit (AIDK), which arms to provide a unified interface to end user to control Dolby MS12 function in android system. Dolby Audio manager will also be enabled based on AIDK.
Dolby MS12 product level certification will include named ASDK test from Sept,2023 based on Android ATV system.

To support AIDK in your android ATV system, the release package will include a zip file named "Dolby-Android-AIDK.zip". please unzip the named "dolby" folder to your android folder (Android-s/vendor/) before compiling your android image. The final folder location:

```
droid12:/mnt/fileroot2/ /android-s/vendor$ tree -L 1 -A
.
├── amlogic
├── dolby
├── google
├── google_gtvs
├── google_gtvs_gtv
├── google_vts
├── partner_modules
├── widevine
└── wifi_driver
```

Currently, AIDK is only supported in Amlogic Android 12 ATV system.